Instrumental Variables and Natural Experiments

Eleven examples from the applied literature, sorted by whether the instrument comes from a natural experiment

1. The distinction

A natural experiment is a situation in which some external process, not the researcher and not the units being studied, assigns treatment (or something that shifts treatment) in a way that is as good as random with respect to potential outcomes. An instrumental variable is the variable that enters the first stage of a two-stage estimator. The two are not competing labels: the natural experiment is the source of variation (the design), and IV is one estimator you might apply to it.

In a lottery study, the lottery mechanism is the natural experiment. The lottery number or offer is the instrument. The same natural experiment could instead be analyzed by a simple comparison of means (intent-to-treat), by IV (to recover the effect on compliers), or by difference-in-differences if the shock has a before-and-after structure.

Angrist and Krueger (2001) argue that the best instruments are those backed by a natural experiment, and that weak or contested instruments are usually the ones with no credible natural-experiment story behind them. Dunning (2012) adds a further criterion: the researcher must be able to explain why assignment is as-if random, not merely assert it. The sorting below follows that logic. The test question for each row is the one in the fifth column: who or what assigned the instrument's value, and could that assigner have known anything about the outcome?

2. Group A: instruments that come from a natural experiment

A known, documented device (a lottery, biology, random case rotation) assigned the instrument. Credibility comes from understanding the mechanism rather than from an argument about plausibility.

#	Instrument (study)	Treatment it shifts	Confound / selection it addresses	Who or what assigns the instrument?	Strength
1	Draft lottery number (Angrist 1990)	Military service in Vietnam era	Self-selection into the military on health, ability, risk tolerance, and civilian job prospects	A lottery drawing of birthdates. The researcher can point to the randomizing device.	Strong. Large first stage; exclusion restriction widely accepted.

#	Instrument (study)	Treatment it shifts	Confound / selection it addresses	Who or what assigns the instrument?	Strength
2	Lottery offer for an oversubscribed program (Oregon Health Insurance Experiment, Finkelstein et al. 2012; charter school lotteries, Abdulkadiroglu et al. 2011)	Medicaid enrollment; charter school attendance	Selection into enrollment on motivation, health status, family background	A government or school lottery. Randomized, but not by the researcher.	Strong. Partial compliance (about 30% in Oregon) but large samples; estimates a LATE for compliers.
3	Judge or examiner leniency (Kling 2006; Aizer & Doyle 2015; Dobbie, Goldin & Yang 2018)	Incarceration, pretrial detention, disability benefit receipt	Unobserved defendant characteristics (offense severity, demeanor, history) that drive both the decision and later outcomes	Random or rotational case assignment within a courthouse or office.	Strong first stage. Main concern is monotonicity: a judge lenient on one case type may be harsh on another.
4	Sex composition of first two children (Angrist & Evans 1998)	Having a third child	Parents who want large families may also have weaker labor-market attachment	Biology. Sex at birth is as good as random.	Strong. Precisely estimated in census data even though the fertility shift is modest (6 to 7 pp).
5	Quarter of birth (Angrist & Krueger 1991) [borderline]	Years of schooling, via compulsory attendance laws tied to age	Ability bias	Birth timing interacting with school-entry and leaving-age rules. The mechanism is external and well documented, but season of birth correlates with family characteristics, so as-if-random is contestable.	Weak. The 180-instrument version had F-statistics near 1 (Bound, Jaeger & Baker 1995). The textbook weak-IV cautionary tale.

Note on row 5: Angrist and Krueger treat quarter of birth as a natural experiment because the compulsory-schooling mechanism is external and well understood. Later work showed the as-if-random claim is weaker than it first appeared, and the instrument itself is weak. It is included here as the borderline case that shows a natural-experiment story and a strong first stage are separate requirements.

3. Group B: plausibly exogenous instruments without a natural experiment

Something external produced the variation, but no randomizing device assigned it. The researcher must argue that the variation is unrelated to unobservables, usually by reasoning about the channel through which the instrument affects the treatment. Many methodologists would not call these natural experiments at all.

#	Instrument (study)	Treatment it shifts	Confound / selection it addresses	Who or what assigns the instrument?	Strength
6	Distance to nearest college (Card 1995)	Years of schooling	Ability bias and family background	Family residential choice. Families near colleges differ, and proximity may affect local labor markets directly.	Medium. Adequate first stage; exclusion restriction debated.
7	Settler mortality (Acemoglu, Johnson & Robinson 2001)	Quality of current institutions (expropriation risk)	Reverse causality (rich countries can afford institutions); omitted geography and culture	Historical disease environment. External, but nobody randomized it, and disease may affect income through channels other than institutions.	Medium. Respectable first stage; data fragility (Albouy 2012) and exclusion restriction heavily contested.
8	Bartik / shift-share (Bartik 1991; Blanchard & Katz 1992)	Local labor demand (employment or wage growth)	Local supply shocks and region-specific confounds	National industry trends weighted by a region's initial industry mix. Identification rests on the initial shares being exogenous (Goldsmith-Pinkham, Sorkin & Swift 2020).	Strong first stage; exclusion restriction depends on the shares.
9	Rainfall shocks (Miguel, Satyanath & Sergenti 2004)	Economic growth in agrarian African countries	Reverse causality between growth and civil conflict; omitted institutional quality	Weather. Exogenous to politics, but rainfall may affect conflict directly through terrain, mobility, or disease.	Weak to medium. Marginal first stage in some specifications.
10	Electoral cycles in police hiring (Levitt 1997)	Police force size	Simultaneity: cities hire more police when crime rises	Election timing is institutional, but the hiring response is a behavioral choice by mayors, not an external device.	Weak. Marginal first stage; McCrary (2002) found a coding error, after which estimates were very imprecise.
11	State political composition and congressional committee representation (Payne 1998; Andreoni & Payne 2003)	Government grants received by nonprofits	Grant allocation and private donations both respond to local demand for the nonprofit's services	Legislative politics. Political composition could affect donors' giving directly.	Medium to weak. Exclusion restriction debatable.

4. Where the line is drawn

There is no sharp consensus, but three distinctions come up repeatedly.

- **RCT versus natural experiment.** The difference is who controlled the randomization. The Oregon lottery was run by the state, not by researchers, so it is a natural experiment even though it behaves like an RCT. Some authors call this a randomized natural experiment.
- **Natural experiment versus quasi-experiment.** This is the fuzziest boundary. Difference-in-differences, regression discontinuity, and synthetic control are usually called quasi-experimental. A workable rule: a natural experiment makes a claim about as-if random assignment; a quasi-experiment makes a

claim about comparability of trends or of units near a cutoff (parallel trends, continuity).

- **Natural experiment versus IV.** IV is an estimator; the natural experiment is a design. A strong instrument is almost always one with an identifiable natural experiment behind it. The Group B instruments can still be valid, but their validity rests on an argument rather than on a mechanism.

5. Teaching heuristic

Strength tracks how the variation is generated. Lotteries and random assignment (rows 1 to 4) are strong because a device did the assigning. Geographic proximity and historical conditions (rows 6 to 8) have decent first stages but leaky exclusion restrictions. Calendar, weather, and behavioral-timing instruments (rows 5, 9, 10) are often weak because the shift in treatment is small or the assigner is not external. A useful exercise is to have students identify the assigner for each instrument before looking at the strength rating, then compare.

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